



- **Core Objective:** The focus shifts to analyzing scenarios that involve drawing statistical inferences about two distinct binomial parameters of interest, designated as p_1 and p_2 .
- **Methodological Framework:** The presented analytical techniques utilize large-sample hypothesis testing and confidence interval (CI) procedures.
- **Underlying Mathematical Principle:** These statistical procedures are built upon and justified by the normal approximation to the binomial distribution.

- **Sampling Framework:** The procedure utilizes two independent random samples of sizes n_1 and n_2 taken from two separate populations.
- **Observed Counts:** Let x_1 and x_2 represent the number of observed successes or items belonging to the class of interest in samples 1 and 2, respectively.
- **Normal Approximation:** By applying the normal approximation to the binomial distribution for both populations, the estimators of the population proportions, $\hat{P}_1 = X_1/n_1$ and $\hat{P}_2 = X_2/n_2$, are treated as having approximately normal distributions.

- **Hypotheses to Test:** The analysis is structured to evaluate the equality of the two population proportions using the following statistical statements:
 - Null Hypothesis $H_0: p_1 = p_2$
 - Alternative Hypothesis $H_1: p_1 \neq p_2$

- **Standardized Statistics:** The difference between two independent sample proportions can be standardized into a random variable Z that follows an approximately standard normal distribution, $N(0, 1)$:

$$Z = \frac{\hat{P}_1 - \hat{P}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \quad (5-23)$$

- **Null Hypothesis Foundation:** Under the assumption that the null hypothesis $H_0: p_1 = p_2 = p$ is true, the population proportions are equal, making the true difference term $(p_1 - p_2)$ equal to zero.

- **Pooled Estimator:** Since the common parameter p is unknown, it is estimated by pooling the data from both samples using the following estimator:

$$\hat{P} = \frac{X_1 + X_2}{n_1 + n_2}$$

- **Operational Test Statistic:** Substituting the pooled estimate $\hat{P}$ into the standard error of equation (5-23) establishes the final test statistic Z_0 for evaluating the hypothesis:

$$Z_0 = \frac{\hat{P}_1 - \hat{P}_2}{\sqrt{\hat{P}(1 - \hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

• Testing Hypotheses on the Equality of Two Binomial Proportions:

Null hypothesis: $H_0: p_1 = p_2$.

- **Operational Test Statistic:** The quantitative calculation relies on standardizing the difference in sample proportions using a pooled sample proportion $\hat{p}$:

$$z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \quad (5-24)$$

- **Testing Protocol:** The statistical framework maps out the designated **Alternative Hypotheses**, **Approximate P-Value**, and **Rejection Criterion** for **Approximately Fixed-Level Tests** across three distinct configurations:
 - **Two-Sided Setup:**
 - **Alternative Hypotheses:** $H_1: p_1 \neq p_2$
 - **Approximate P-Value:** $2[1 - \Phi(|z_0|)]$
 - **Rejection Criterion for Approximately Fixed-Level Tests:** $|z_0| > z_{\alpha/2}$

- **Upper-Tailed Setup:**
 - **Alternative Hypotheses:** $H_1: p_1 > p_2$
 - **Approximate P-Value:** $1 - \Phi(z_0)$
 - **Rejection Criterion for Approximately Fixed-Level Tests:** $z_0 > z_\alpha$

- **Lower-Tailed Setup:**
 - **Alternative Hypotheses:** $H_1: p_1 < p_2$
 - **Approximate P-Value:** $\Phi(z_0)$
 - **Rejection Criterion for Approximately Fixed-Level Tests:** $z_0 < -z_\alpha$

- **EXAMPLE 5-12 Interocular Lenses:** This study evaluates whether two different types of polishing solutions differ in their impact on manufacturing interocular lenses, using independent samples of 300 lenses per solution.

Solution.

- **1. Parameter of interest:** The parameters of interest are p_1 and p_2 , the true proportions of satisfactory lenses after tumble-polishing with fluid 1 and fluid 2, respectively.
- **2. Null hypothesis, H_0 :** $p_1 = p_2$.

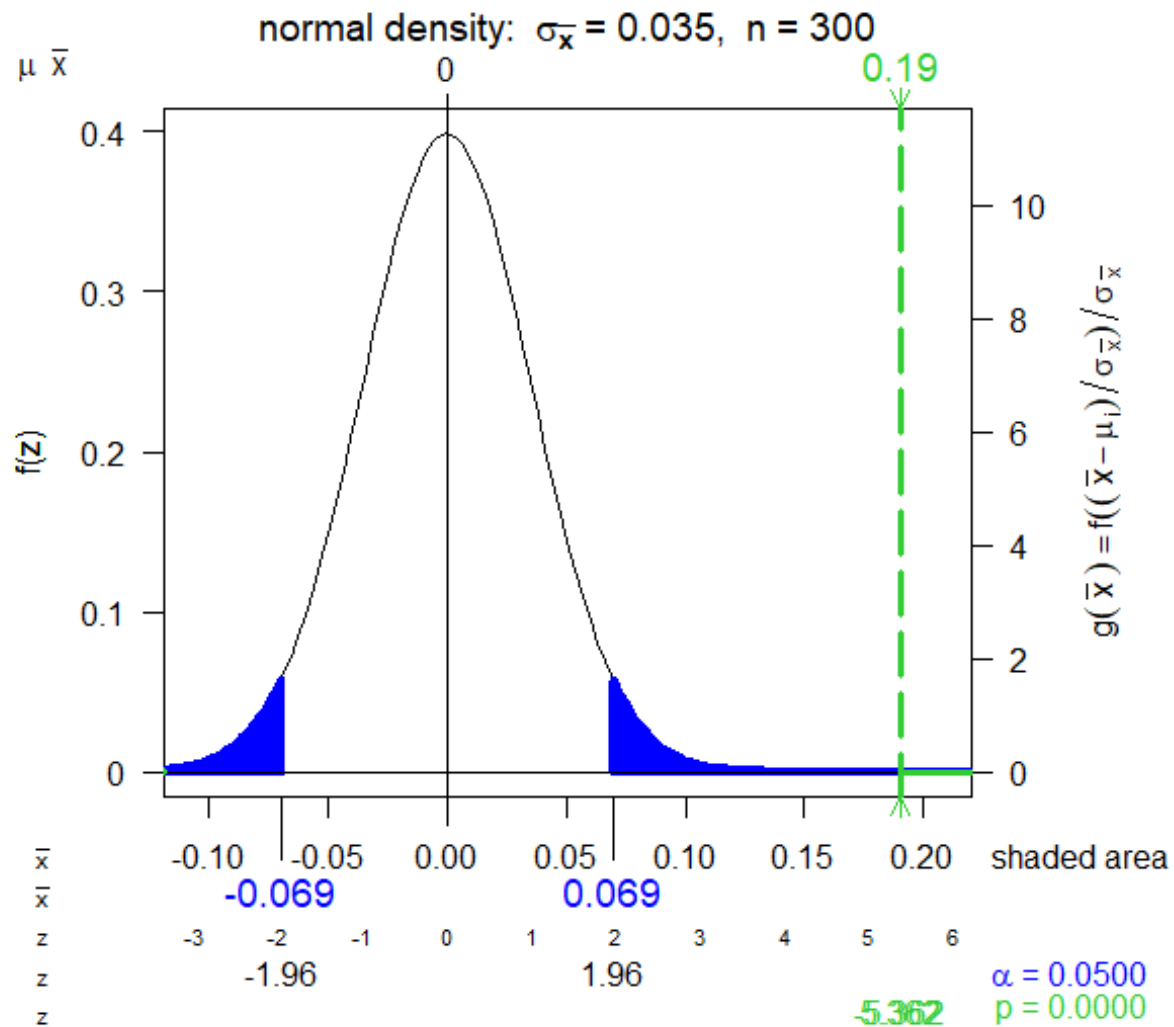
- **3. Alternative hypothesis, $H_1: p_1 \neq p_2$.**
- **4. Test statistic:** The operational formula is $z_0 = (\hat{p}_1 - \hat{p}_2) / \sqrt{\hat{p}(1 - \hat{p})(1/n_1 + 1/n_2)}$, where $\hat{p}_1 \approx 0.8433$, $\hat{p}_2 \approx 0.6533$, $n_1 = n_2 = 300$, and the pooled estimator is $\hat{p} = (253 + 196)/(300 + 300) \approx 0.7483$.
- **5. Reject H_0 if:** Reject the null hypothesis if an approximate p-value is less than 0.05.

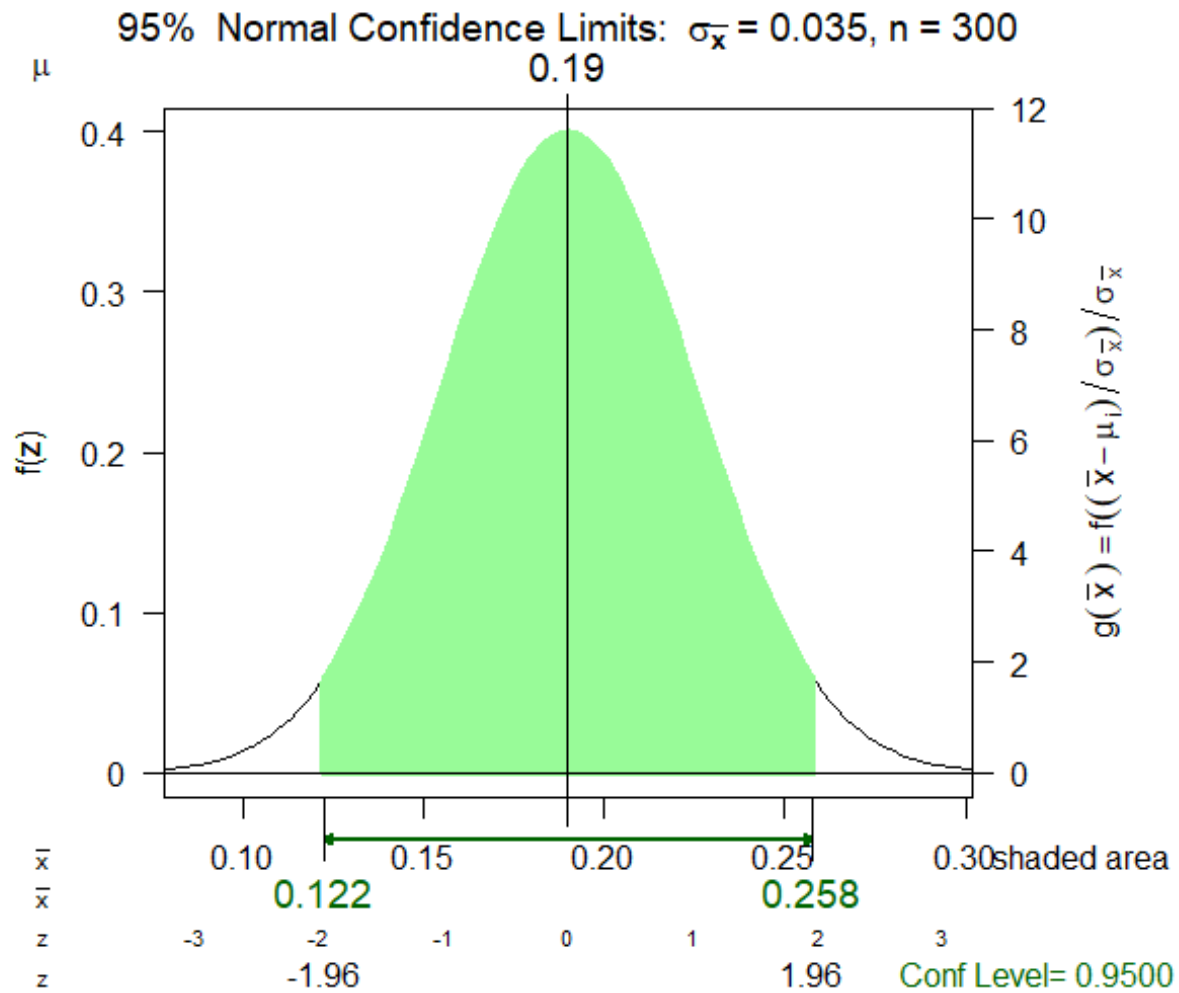
- **6. Computations:** Substituting the empirical values yields a standardized value of:

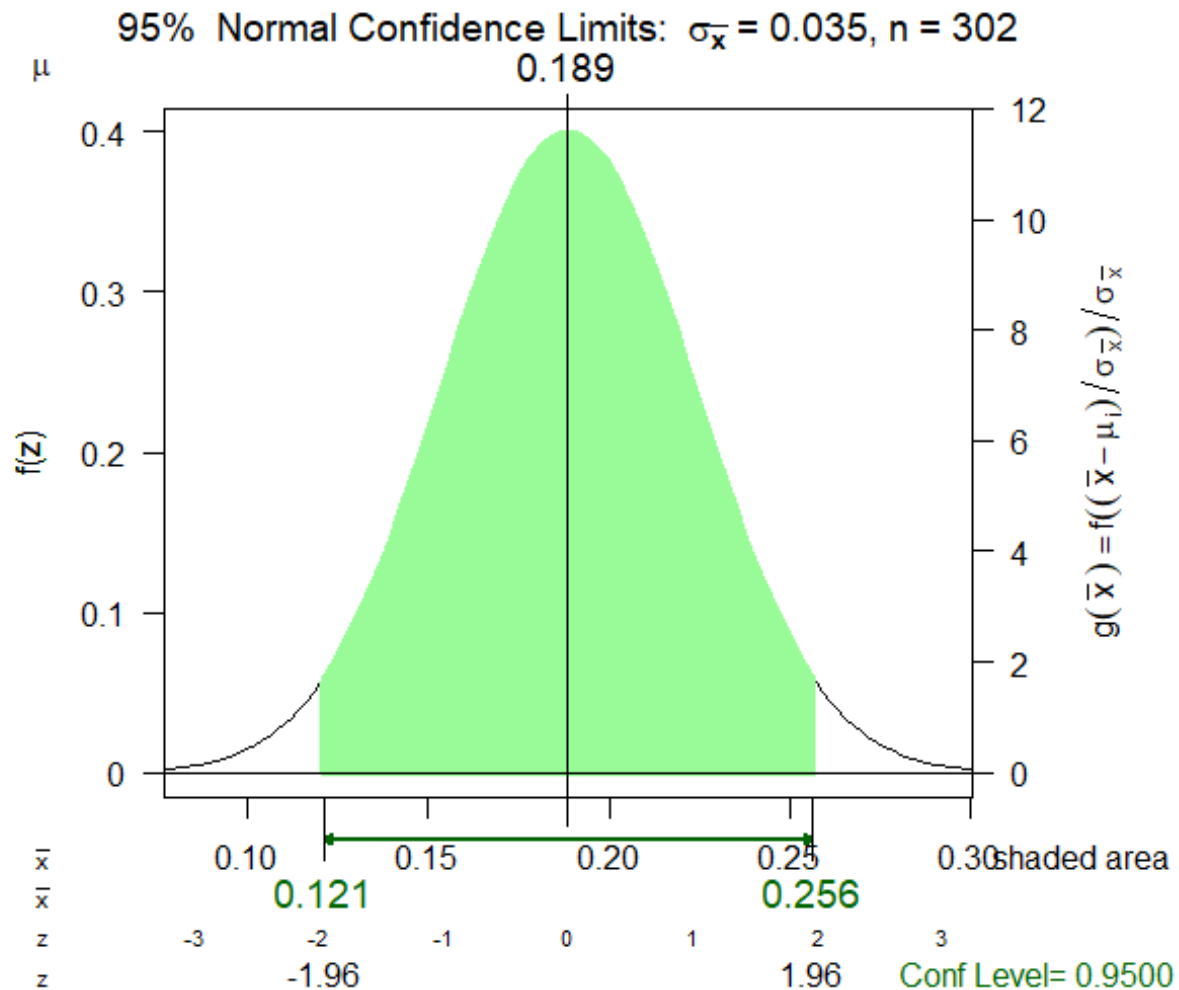
$$z_0 = \frac{253/300 - 196/300}{\sqrt{\frac{253+196}{300+300} \left(1 - \frac{253+196}{300+300}\right) \left(\frac{1}{300} + \frac{1}{300}\right)}} \approx 5.3621$$

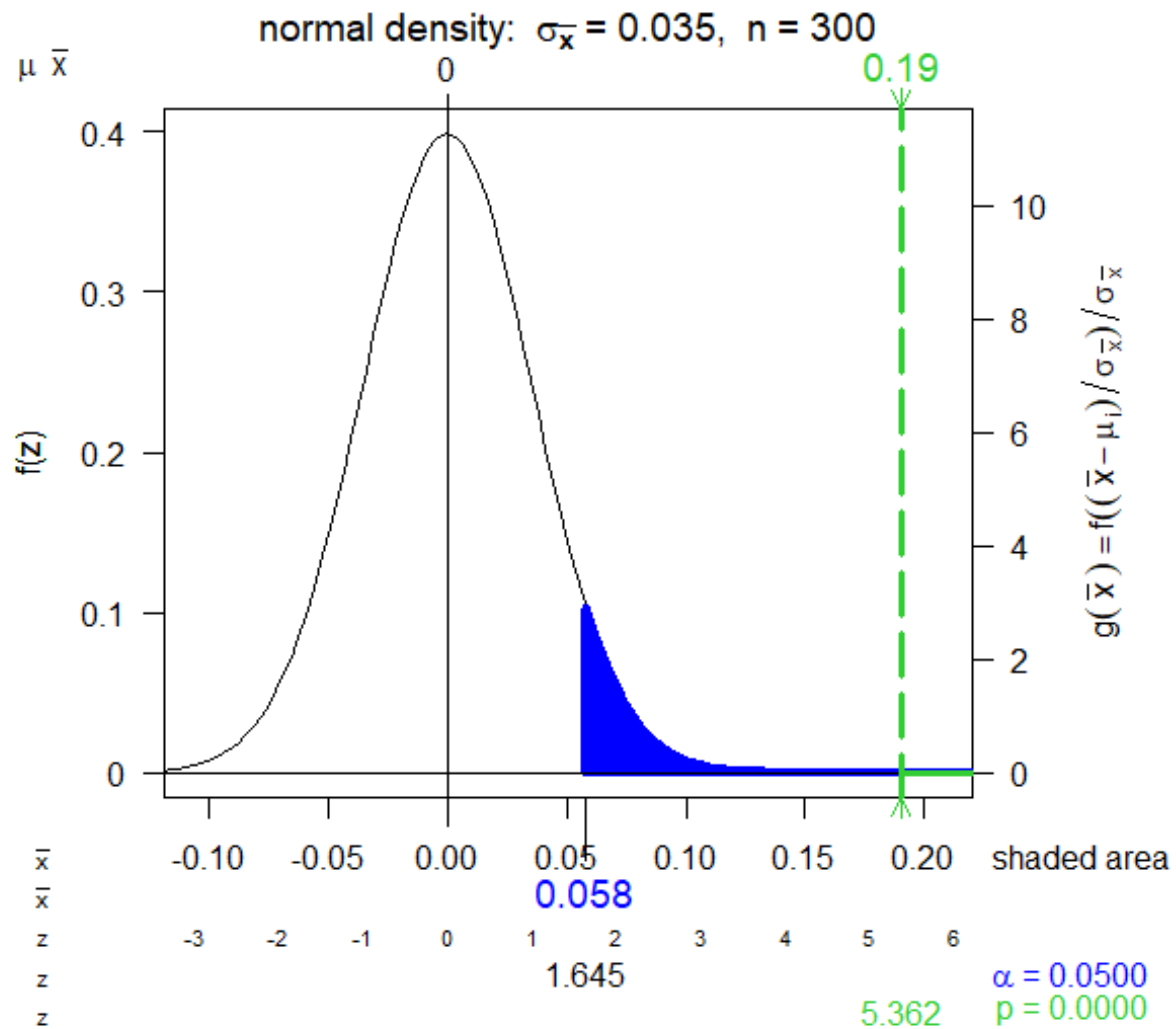
- **7. Conclusions:** Because $z_0 \approx 5.3621$, an approximate p-value is $2 \cdot [1 - \Phi(5.3621)] \approx 8.22 \times 10^{-8}$. Since this value is well below 0.05, we reject the null hypothesis.

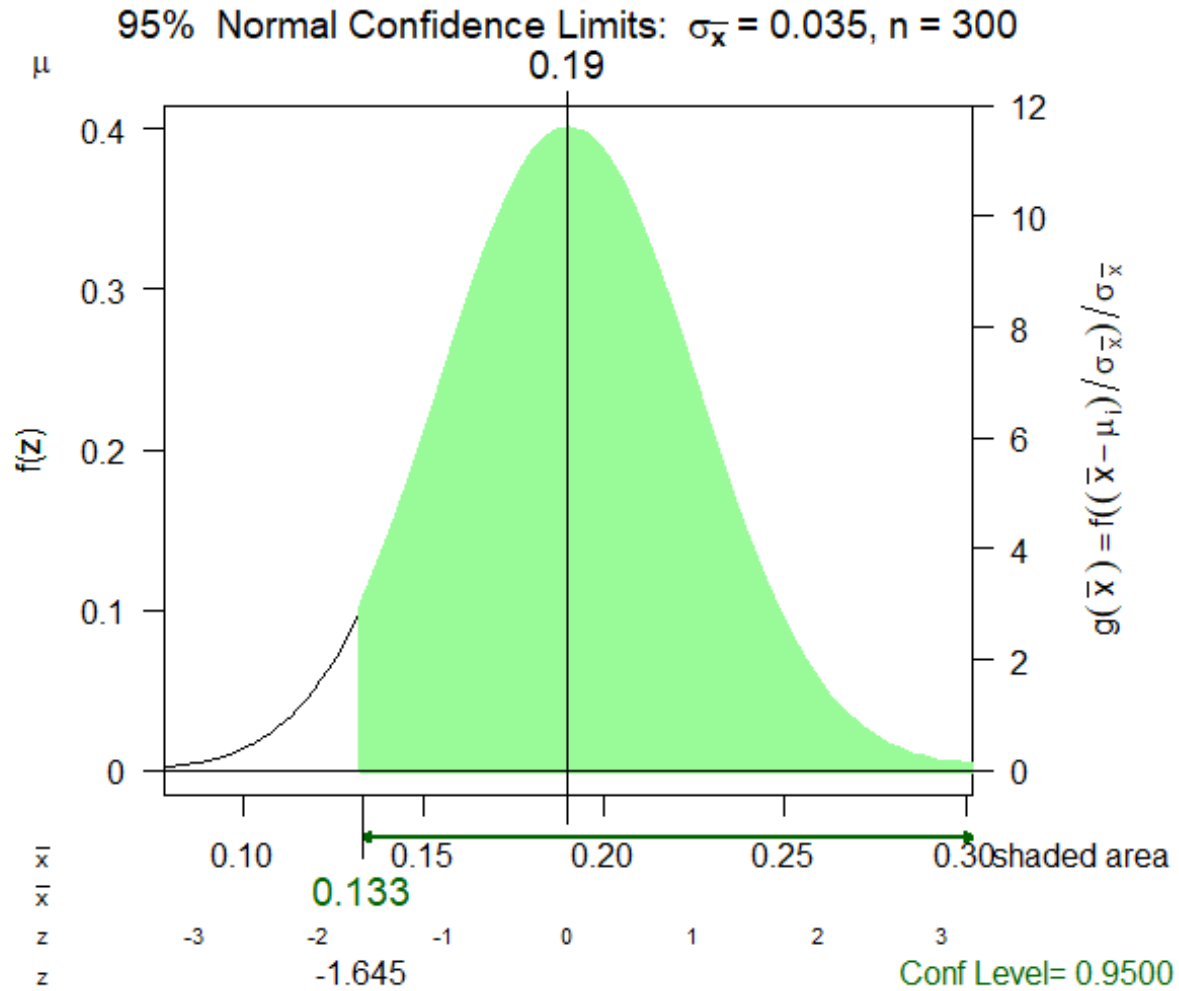
- **Practical engineering conclusion:** There is strong evidence that the two polishing fluids perform differently. Fluid 1 is statistically superior because it produces a significantly higher fraction of nondefective lenses.

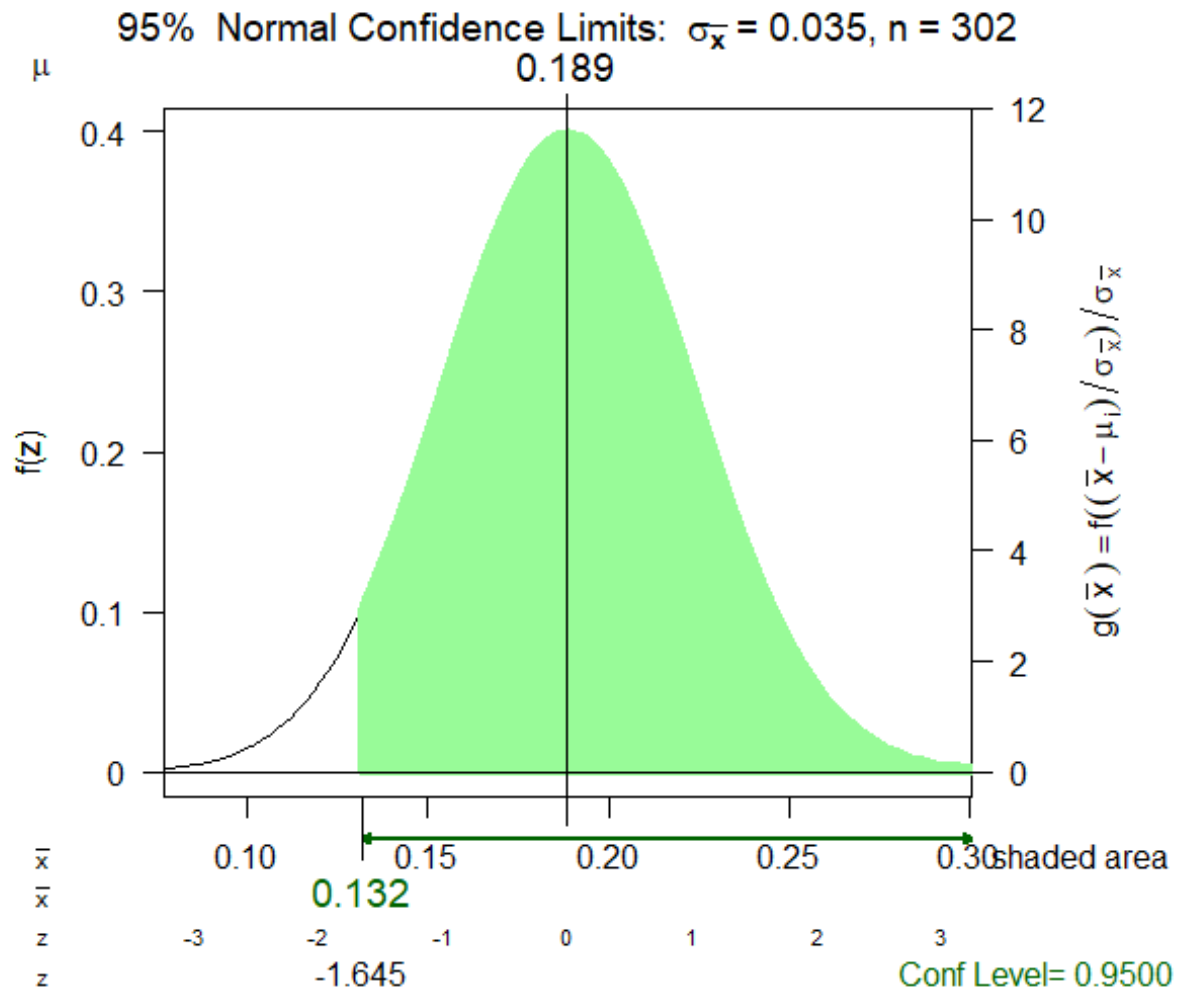


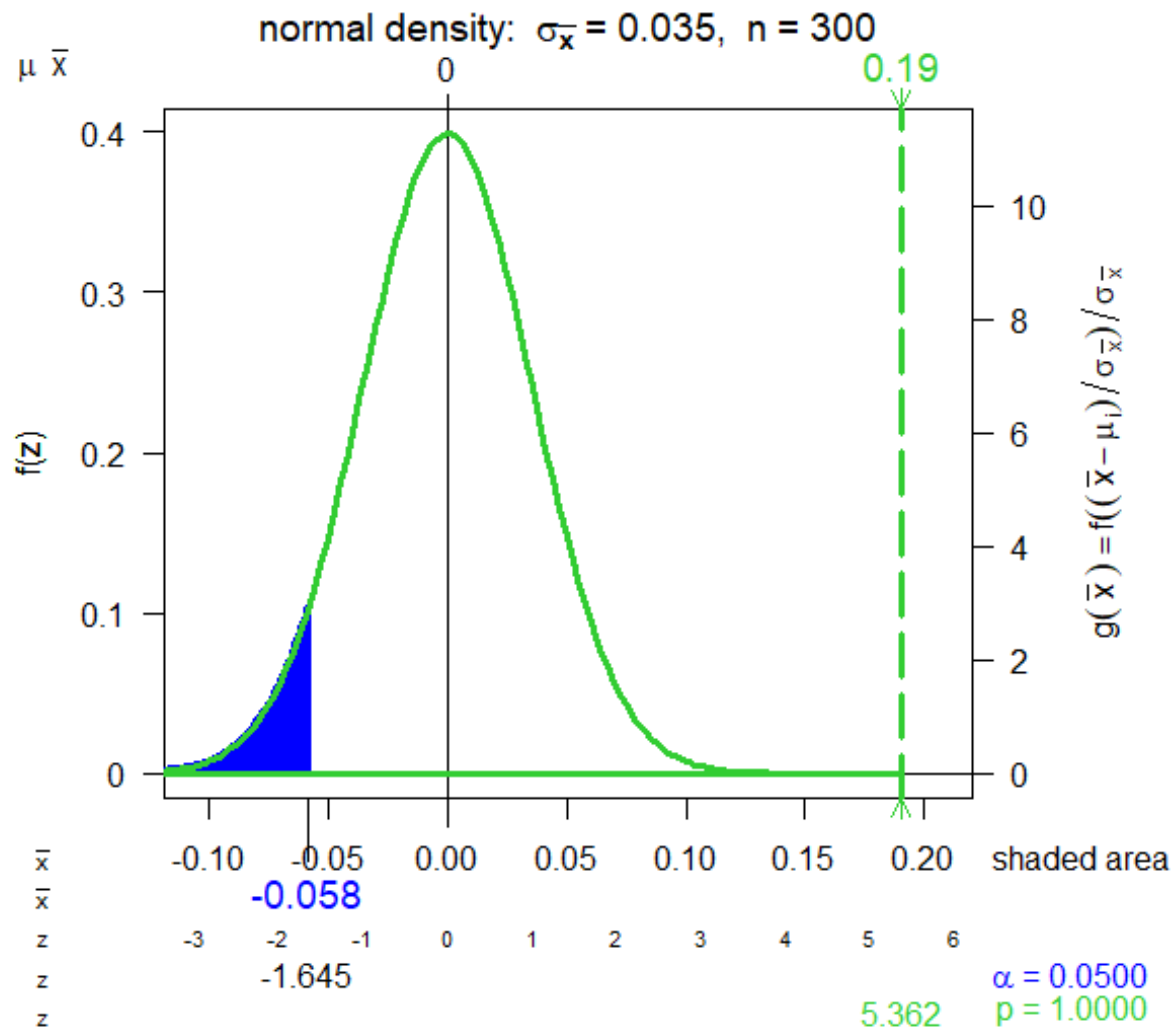


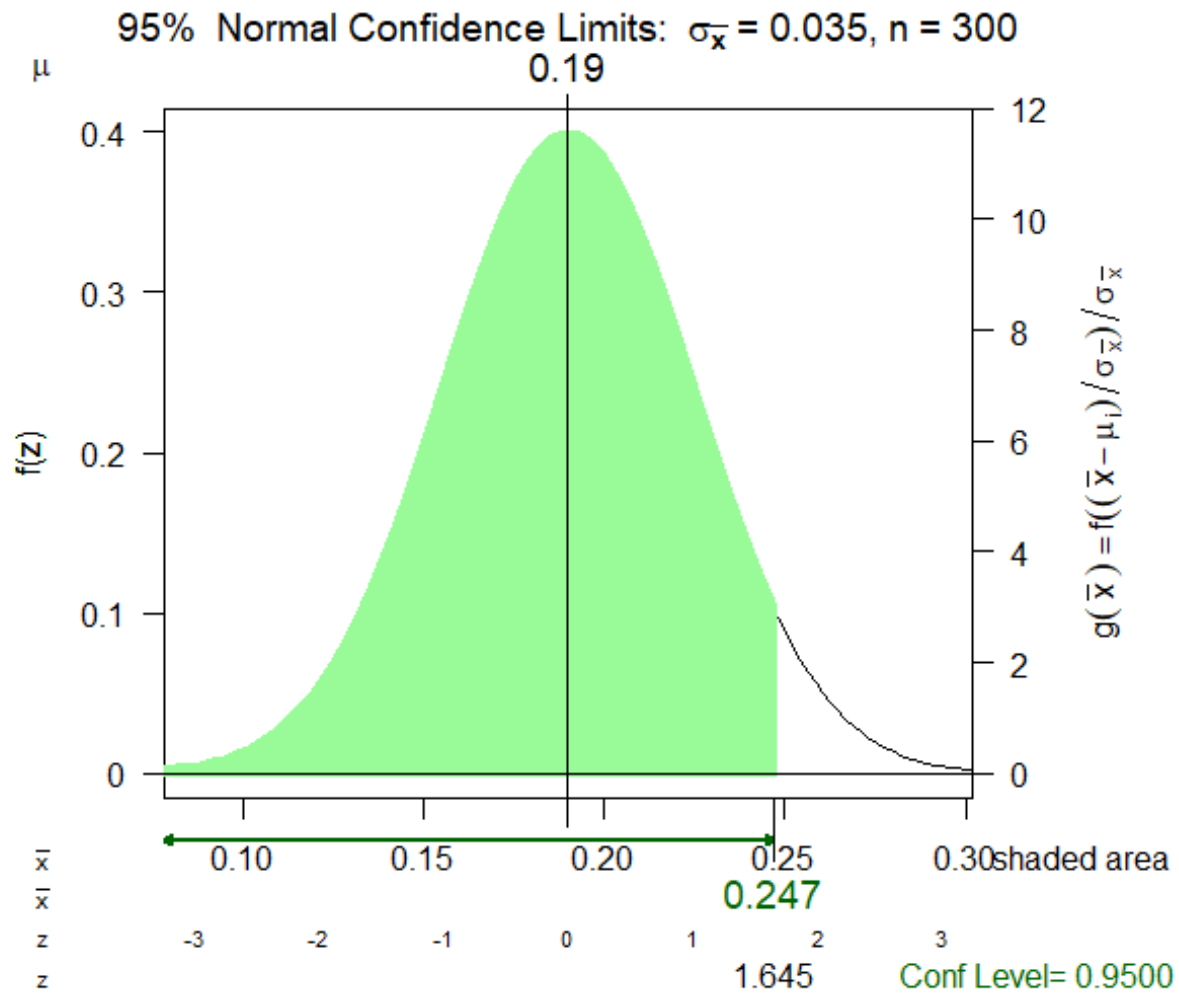


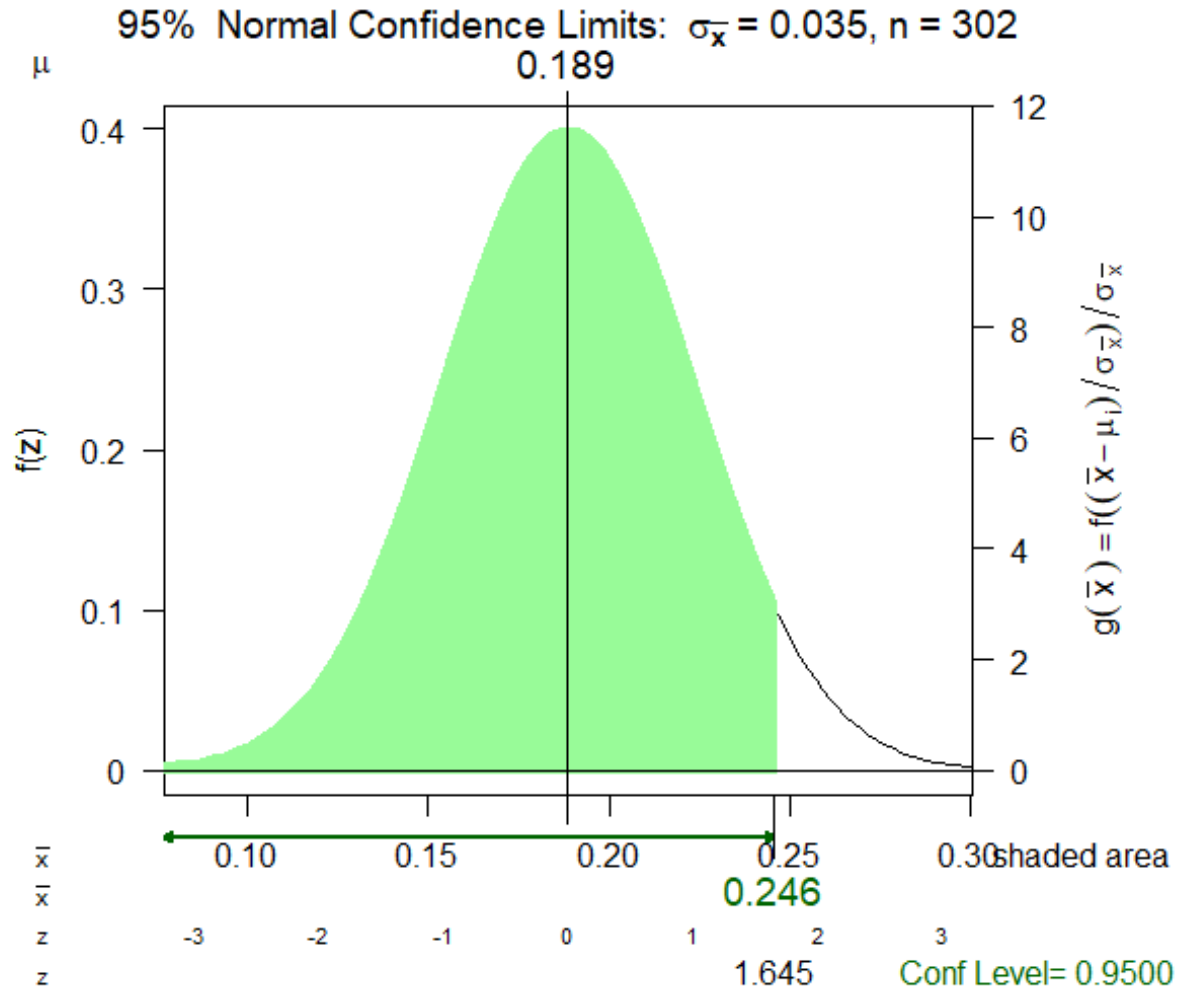












5-6.2 Type II Error and Choice of Sample

- **Complexity in β -Error Estimation:** Computing the β -error for the two-proportion test is more complicated than the single-sample case because the denominator of z_0 assumes equal population proportions ($p_1 = p_2 = p$).
- **True Standard Error:** When the null hypothesis is false ($H_0: p_1 = p_2$ is not true), the population proportions are unequal, meaning the true standard deviation or standard error of $\hat{P}_1 - \hat{P}_2$ is defined by the following expression:

$$\sigma_{\hat{P}_1 - \hat{P}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} \quad (5-25)$$

- **Impact on Power Analysis:** Because this true standard error formula uses distinct values for p_1 and p_2 , calculating power or sample size requirements requires explicit specification of both individual true proportions.

- **Formula Definition:** This framework provides the specific mathematical calculation to evaluate the **β -Error: Two-Sided Difference** in **Proportions Test** under a two-sided alternative hypothesis.

- **The β -Error Approximation:** The probability of committing a Type II error (β) is approximated using the cumulative standard normal distribution function Φ , split into two lines for readability:

$$\beta \approx \Phi \left(\frac{z_{\alpha/2} \sqrt{\bar{p}\bar{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} - (p_1 - p_2)}{\sigma_{\hat{P}_1 - \hat{P}_2}} \right) - \Phi \left(\frac{-z_{\alpha/2} \sqrt{\bar{p}\bar{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} - (p_1 - p_2)}{\sigma_{\hat{P}_1 - \hat{P}_2}} \right) \quad (5-26)$$

- **Weighted Parameter Components:** The underlying calculations rely on the pooled true proportions across both samples:

- $\bar{p} = \frac{n_1 p_1 + n_2 p_2}{n_1 + n_2}$

- $\bar{q} = 1 - \bar{p} = \frac{n_1(1-p_1) + n_2(1-p_2)}{n_1 + n_2}$

- **Standard Error Input:** The denominator term $\sigma_{\hat{P}_1 - \hat{P}_2}$ represents the true standard error under the alternative hypothesis, which is pulled directly from equation 5-25.

- **Formula Application:** This section outlines how to calculate the β -Error: One-Sided Difference in Proportions Test depending on the direction of the directional alternative hypothesis.
- **Upper-Tailed Scenario:** If the alternative hypothesis specifies that the first proportion is greater than the second ($H_1: p_1 > p_2$), the β -error is approximated by:

$$\beta \approx \Phi \left(\frac{z_\alpha \sqrt{\bar{p}\bar{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} - (p_1 - p_2)}{\sigma_{\hat{P}_1 - \hat{P}_2}} \right) \quad (5-27)$$

- **Lower-Tailed Scenario:** If the alternative hypothesis specifies that the first proportion is less than the second ($H_1: p_1 < p_2$), the β -error calculation changes to:

$$\beta \approx 1 - \Phi \left(\frac{-z_\alpha \sqrt{\bar{p}\bar{q} \left(\frac{1}{n_1} + \frac{1}{n_2} \right)} - (p_1 - p_2)}{\sigma_{\hat{P}_1 - \hat{P}_2}} \right) \quad (5-28)$$

- **Component Definitions:** In both mathematical models, the values for $\bar{p}$, $\bar{q}$, and $\sigma_{\hat{P}_1 - \hat{P}_2}$ are derived uniformly using the identical pooled parameters established for the two-sided condition.

- **Objective:** This formula determines the identical required sample size ($n_1 = n_2 = n$) to achieve an approximate significance level of α and ensure the Type II error (β) remains below a designated maximum target threshold (β_0).
- **Sample Size for a Two-Sided Hypothesis Test on the Difference in Two Binomial Proportions:** For a two-sided alternative hypothesis, the required sample size for each group is calculated as follows:

$$n \approx \left\lceil \frac{\left(z_{\alpha/2} \sqrt{\frac{(p_1 + p_2)(q_1 + q_2)}{2}} + z_{\beta_0} \sqrt{p_1 q_1 + p_2 q_2} \right)^2}{(p_1 - p_2)^2} \right\rceil \quad (5-29)$$

- **Parameter Inputs:** In equation (5-29), the failure proportions are defined as $q_1 = 1 - p_1$ and $q_2 = 1 - p_2$, and the ceiling function rounds the final value up to the nearest whole integer.
- **One-Sided Adjustment:** If a directional or one-sided alternative hypothesis is evaluated instead, the formula is modified by simply replacing the critical value $z_{\alpha/2}$ with z_{α} .

- **Basis of Derivation:** The confidence interval for $p_1 - p_2$ is derived from the fact that the difference between sample proportions, when standardized, follows an approximately standard normal distribution (Z).
- **Approximately Standard Normal Variable:** The variable Z is defined by the difference between the observed and true proportions, divided by the pooled standard error:

$$Z = \frac{\hat{P}_1 - \hat{P}_2 - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

- **Probability Foundation:** The construction uses the standard probability statement $P(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}) \approx 1 - \alpha$.
- **Methodology:** By substituting the expression for Z into the probability statement and isolating the term $p_1 - p_2$, an approximate traditional $100(1 - \alpha)\%$ confidence interval is established.

- **Core Method:** This provides an approximate methodology for calculating the **Traditional Confidence Interval on the Difference in Binomial Proportions**.
- **Data Requirements:** The interval is based on sample proportions ($\hat{p}_1$, $\hat{p}_2$) obtained from two independent random samples of sizes n_1 and n_2 .

- **Confidence Interval Formula:** The approximate $100(1 - \alpha)\%$ confidence interval for the true difference $p_1 - p_2$ is:

$$\hat{p}_1 - \hat{p}_2 - z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \leq p_1 - p_2$$

$$\leq \hat{p}_1 - \hat{p}_2 + z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$
(5-30)

- **Variables in (5-30):** The term $z_{\alpha/2}$ represents the upper $\alpha/2$ percentage point of the standard normal distribution.

- **EXAMPLE 5-13 Crankshaft Bearings:** This study evaluates whether a modification to a surface finishing process improved crankshaft bearing quality by comparing a new sample of $n_2 = 85$ axle shafts (containing 8 defectives) against historical baseline data.

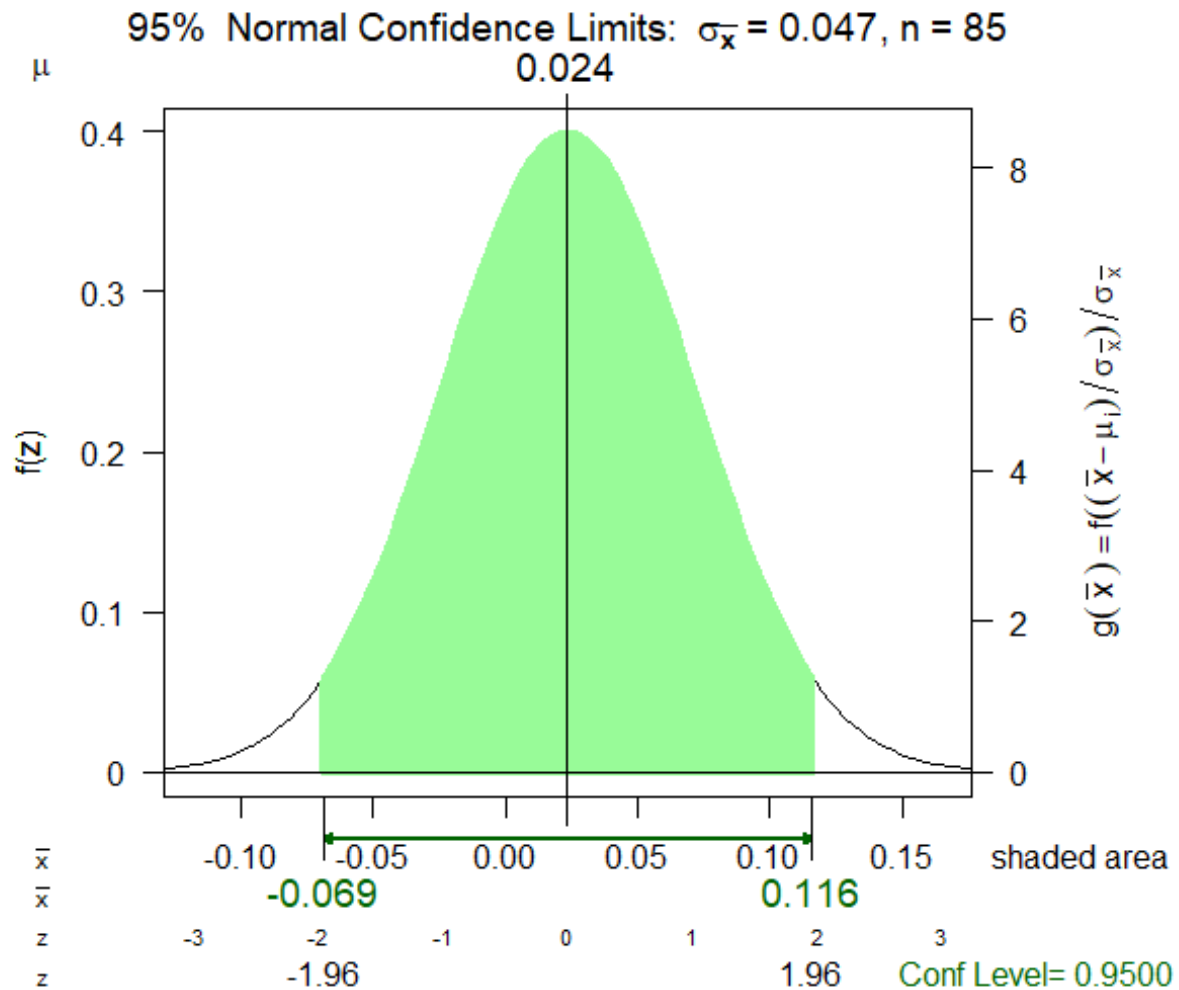
Solution.

- **Sample Parameters:** The analysis establishes proportions from two independent samples where $n_1 = 85$, $\hat{p}_1 = 10/85 \approx 0.1176$, $n_2 = 85$, and $\hat{p}_2 = 8/85 \approx 0.0941$.

- Confidence Interval Calculation:** Substituting these parameters into the formula with an approximately 95% confidence level ($z_{0.025} = \text{qnorm}(0.975)$) yields the following three-line expression with aligned inequality signs:

$$\begin{aligned}
 & 10/85 - 8/85 - \text{qnorm}(0.975) \sqrt{\frac{\frac{10}{85} (1 - \frac{10}{85})}{85} + \frac{\frac{8}{85} (1 - \frac{8}{85})}{85}} \\
 & \leq p_1 - p_2 \\
 & \leq 10/85 - 8/85 + \text{qnorm}(0.975) \sqrt{\frac{\frac{10}{85} (1 - \frac{10}{85})}{85} + \frac{\frac{8}{85} (1 - \frac{8}{85})}{85}} \\
 & -0.0689 \leq p_1 - p_2 \leq 0.1160
 \end{aligned}$$

- **Conclusion:** Because this an approximately 95% confidence interval includes **zero**, the sample data indicates it is unlikely that the process changes have actually reduced the proportion of defective crankshaft bearings being produced.



- **Traditional Method Limitation:** The traditional confidence interval formula in equation 5-30 can exhibit actual confidence levels that deviate substantially from the nominal value (such as a target 95% level).
- **The Agresti-Caffo Adjustment:** To improve coverage accuracy, a simple adjustment can be applied by adding exactly one success and one failure to the empirical data collected from each sample.

- **Adjusted Parameters:** The modified sample sizes and adjusted success proportions are computed as follows:
 - Sample 1: $\tilde{p}_1 = \frac{x_1+1}{n_1+2}$ and $\tilde{n}_1 = n_1 + 2$
 - Sample 2: $\tilde{p}_2 = \frac{x_2+1}{n_2+2}$ and $\tilde{n}_2 = n_2 + 2$
- **Formula Application:** To establish the corrected interval, the standard components $(\hat{p}_1, \hat{p}_2, n_1, n_2)$ are completely replaced by these new adjusted values $(\tilde{p}_1, \tilde{p}_2, \tilde{n}_1, \tilde{n}_2)$ inside equation 5-30.

- **Application to Example 5-13:** The Agresti-Caffo adjustment is applied to the crankshaft bearing data to re-evaluate the difference in binomial proportions.
- **Adjusted Values:** Adding one success and one failure to each sample yields the following modified statistics:
 - Sample 1: $\tilde{p}_1 = \frac{10+1}{85+2} = 0.1264$ and $\tilde{n}_1 = 87$
 - Sample 2: $\tilde{p}_2 = \frac{8+1}{85+2} = 0.1034$ and $\tilde{n}_2 = 87$

- **New Confidence Interval:** Substituting these adjusted metrics into equation 5-30 produces an improved confidence interval of $[-0.0717, 0.1177]$.
- **Interval Comparison:** The adjusted interval length (0.1894) is slightly wider than the traditional interval length (0.1849). This minor increase in width reflects a more precise coverage probability that aligns closer to the nominal 95% level.
- **Final Conclusion Consistency:** Because this new interval also covers zero, the overall engineering conclusion is unchanged, showing no significant difference between the two processes.

